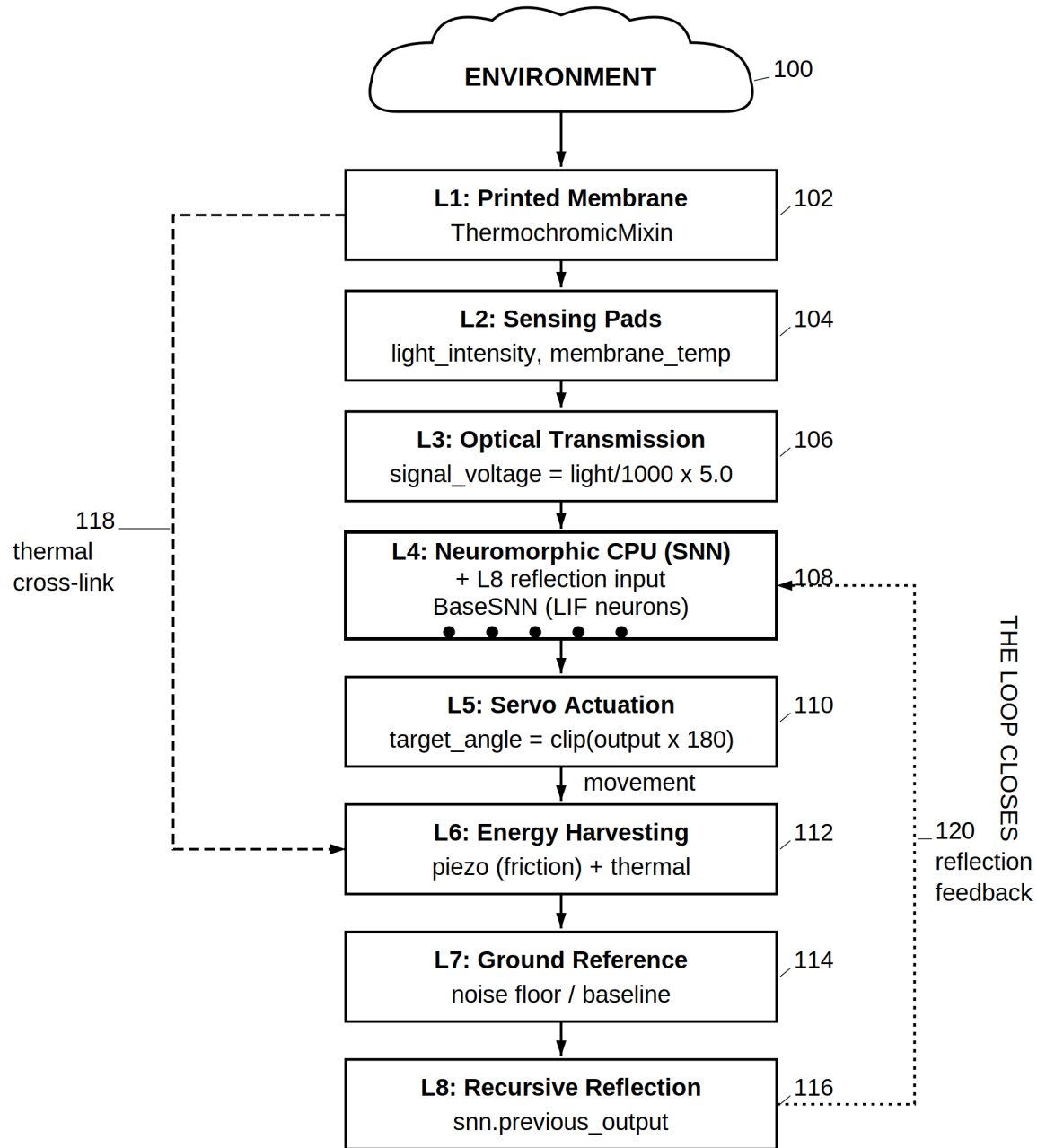


PATENT A

Self-Sustaining Neural-Motor Energy Harvesting Loop■for Autonomous Cognitive Systems

8 drawing sheets

**SIGNAL PATHWAYS:**

- Main spine (L1 through L8 to L4)
- Thermal cross-link (L1 temp to L6)
- Reflection feedback (L8 to L4)

FIG. 1

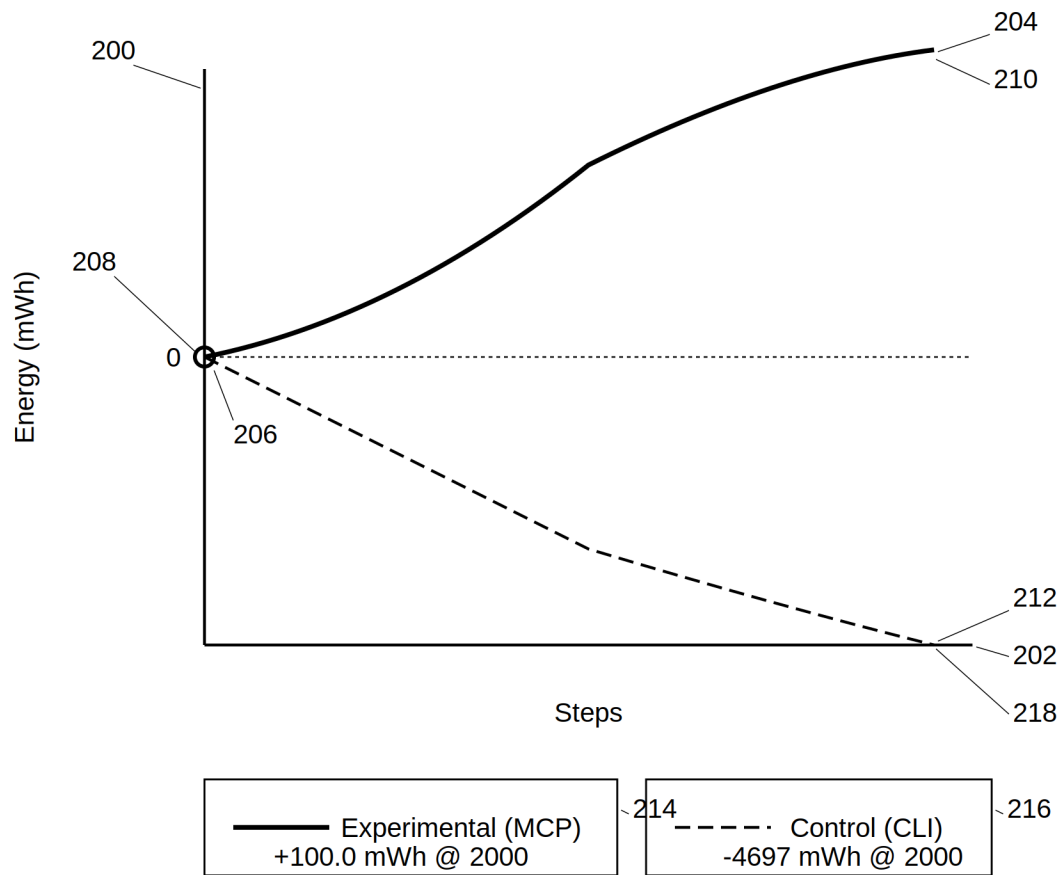


FIG. 2

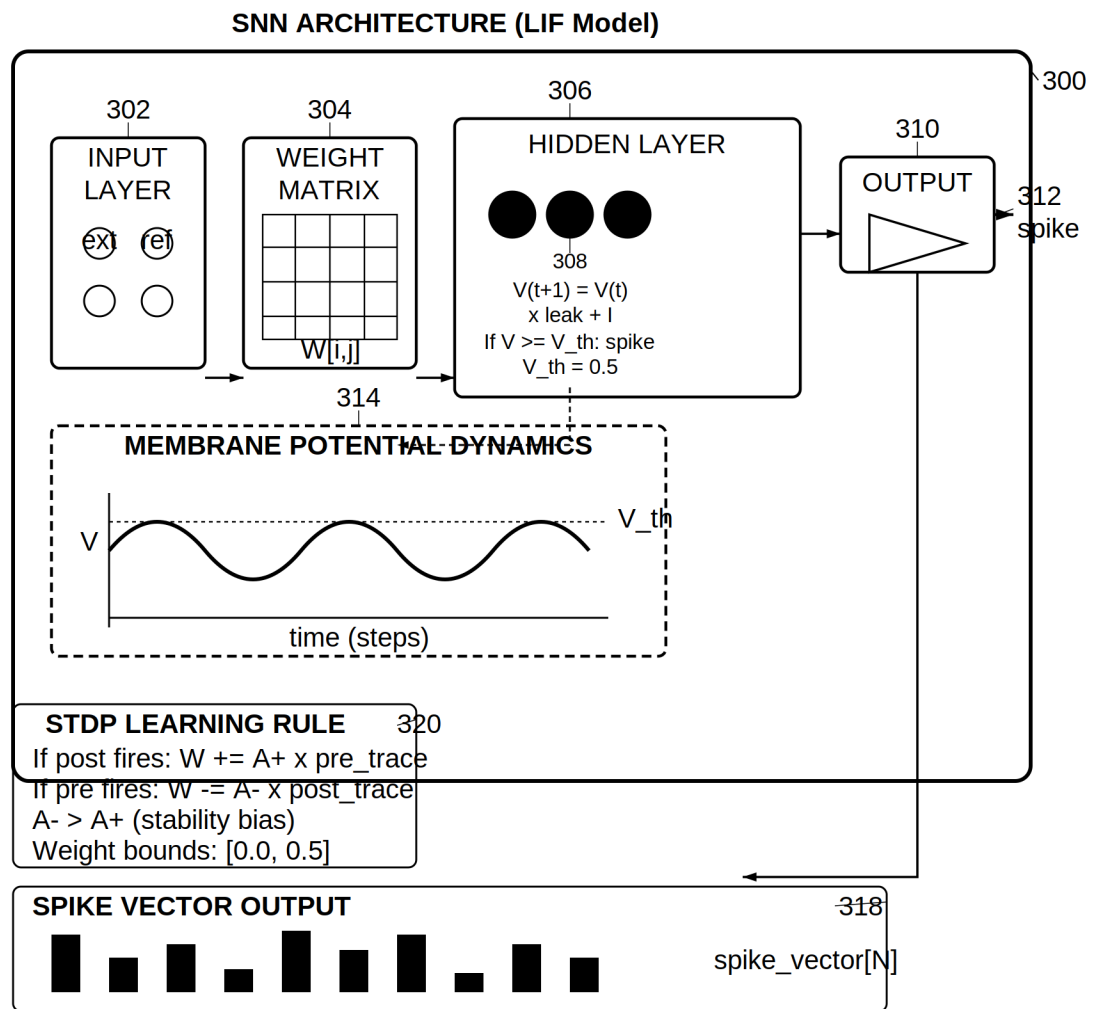


FIG. 3

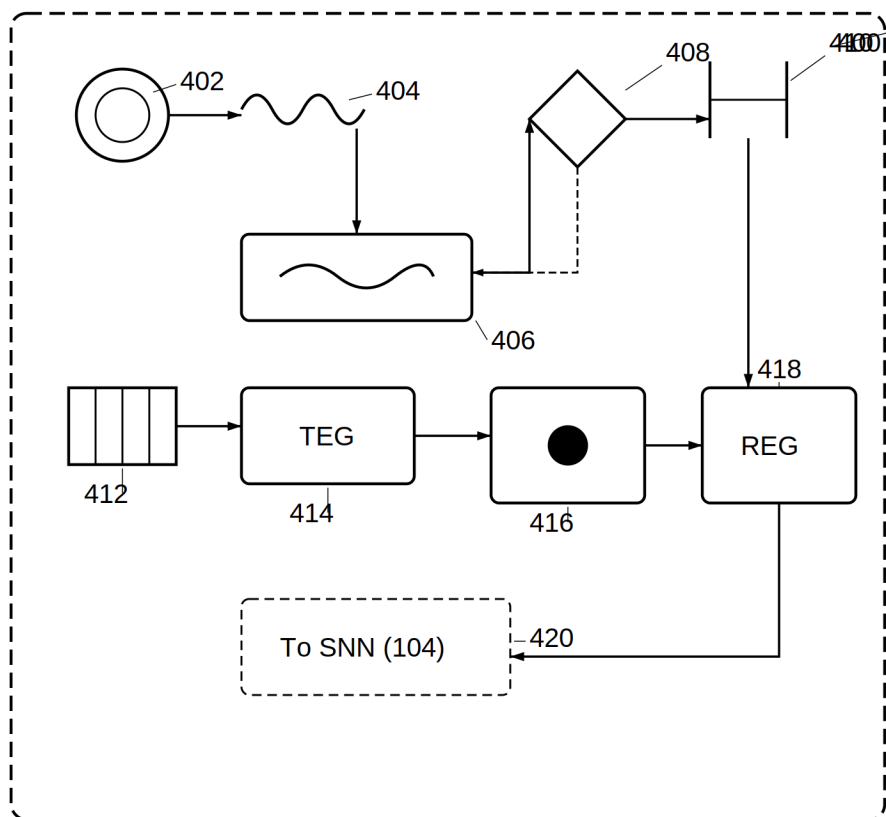


FIG. 4

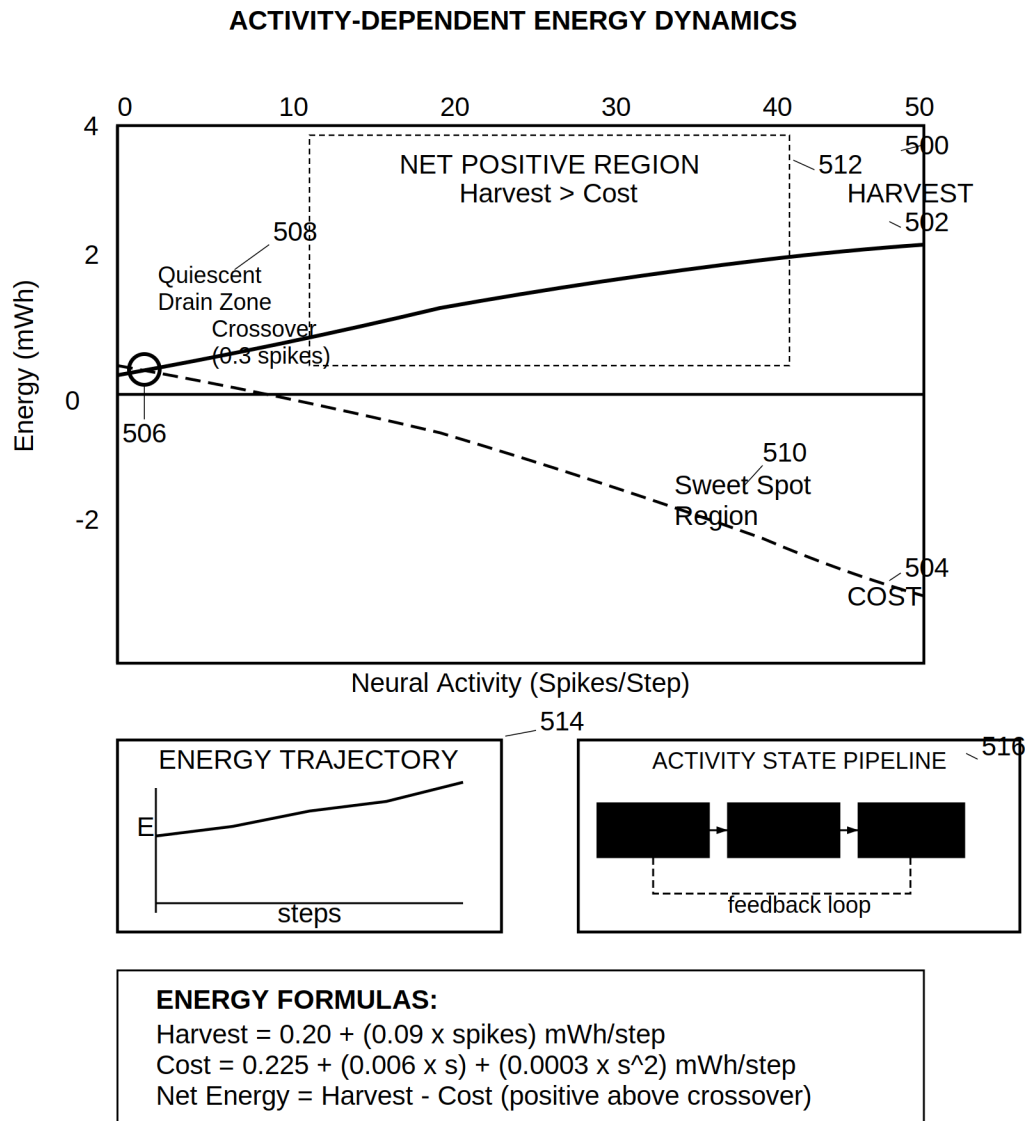
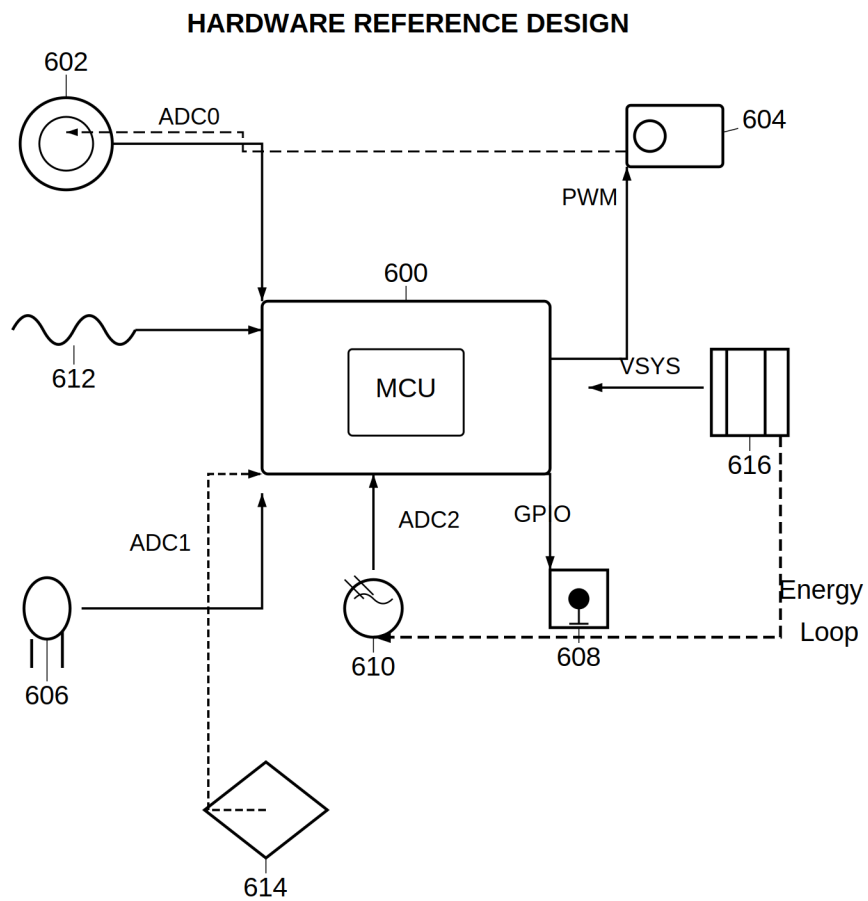


FIG. 5

**FIG. 6**

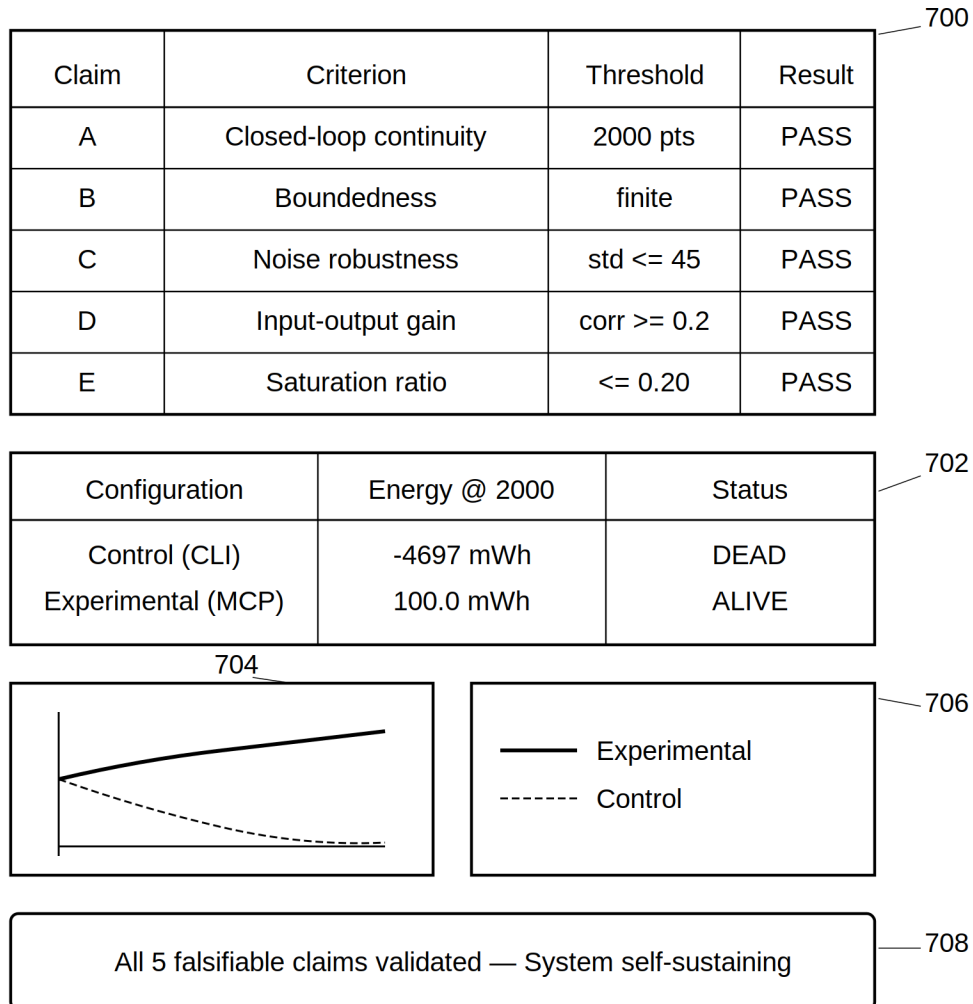


FIG. 7

ENERGY-BOUNDED RECURSIVE CONTROL ARCHITECTURE

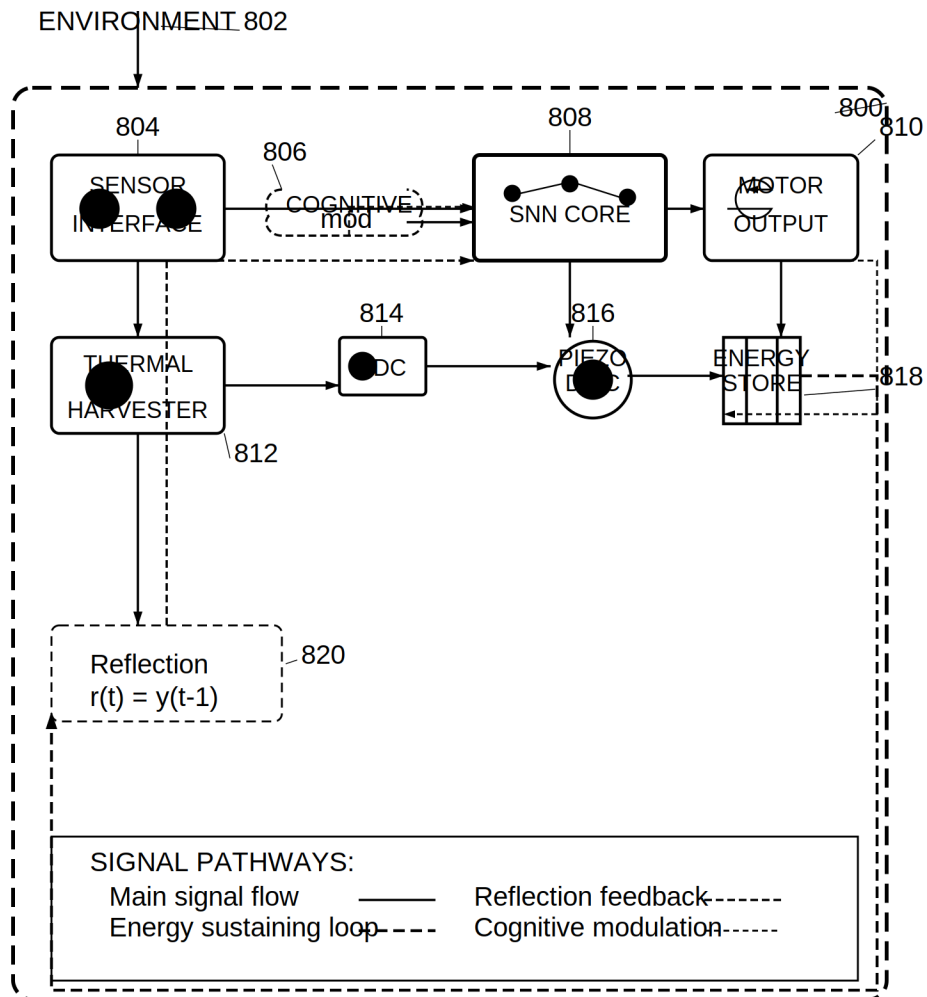


FIG. 8

PATENT B

Recursive Self-Observation Method and System for Spiking Neural Networks with Dynamic Reflection Coefficient

6 drawing sheets

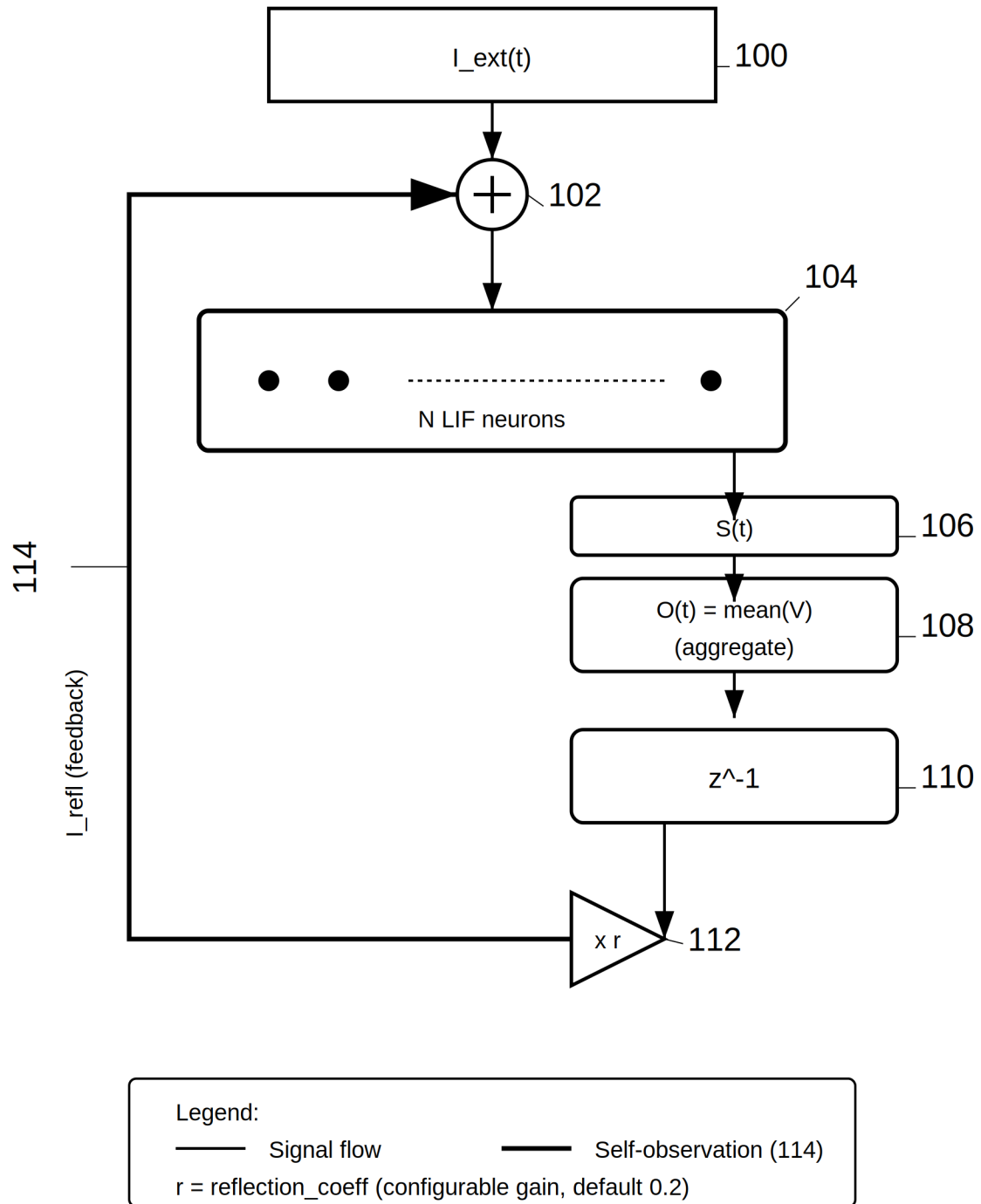
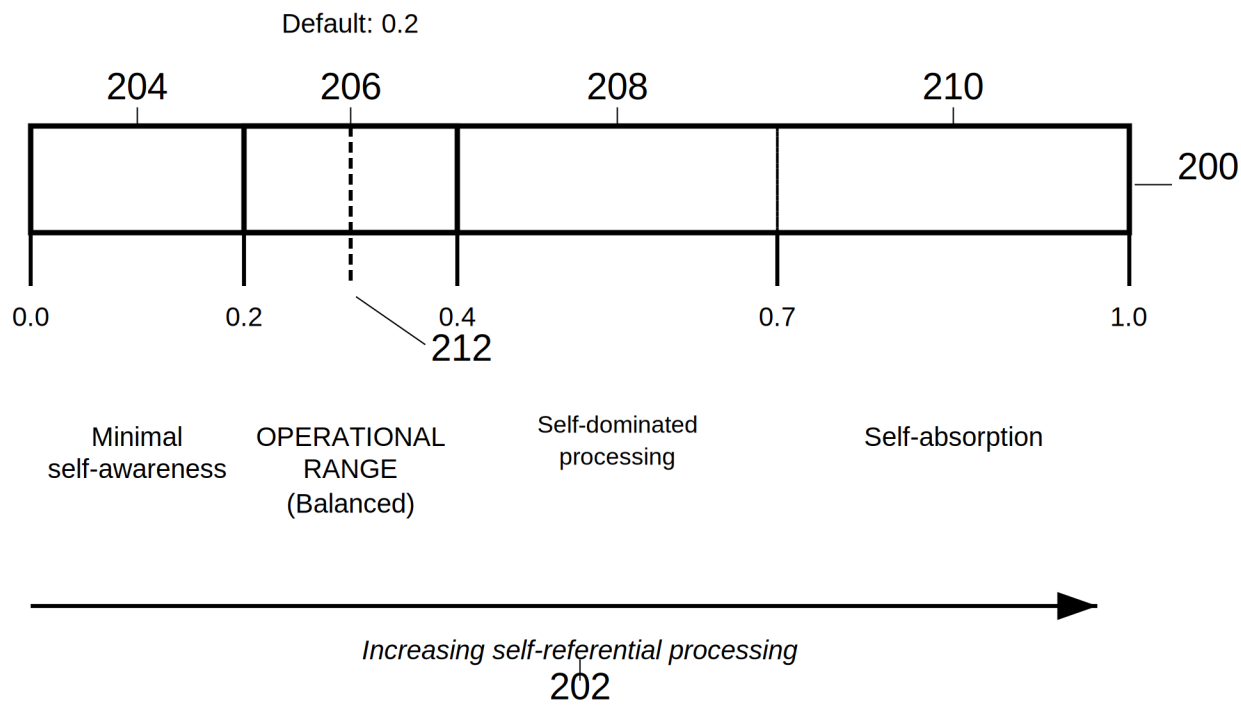


FIG. 1



reflection_coeff = base_coeff + modulation x sensitivity
Default: base_coeff = 0.2, sensitivity = 0.1
Range: 0.1 to 0.3 (within OPERATIONAL zone)
The reflection coefficient is the system's "self-awareness dial"

FIG. 2

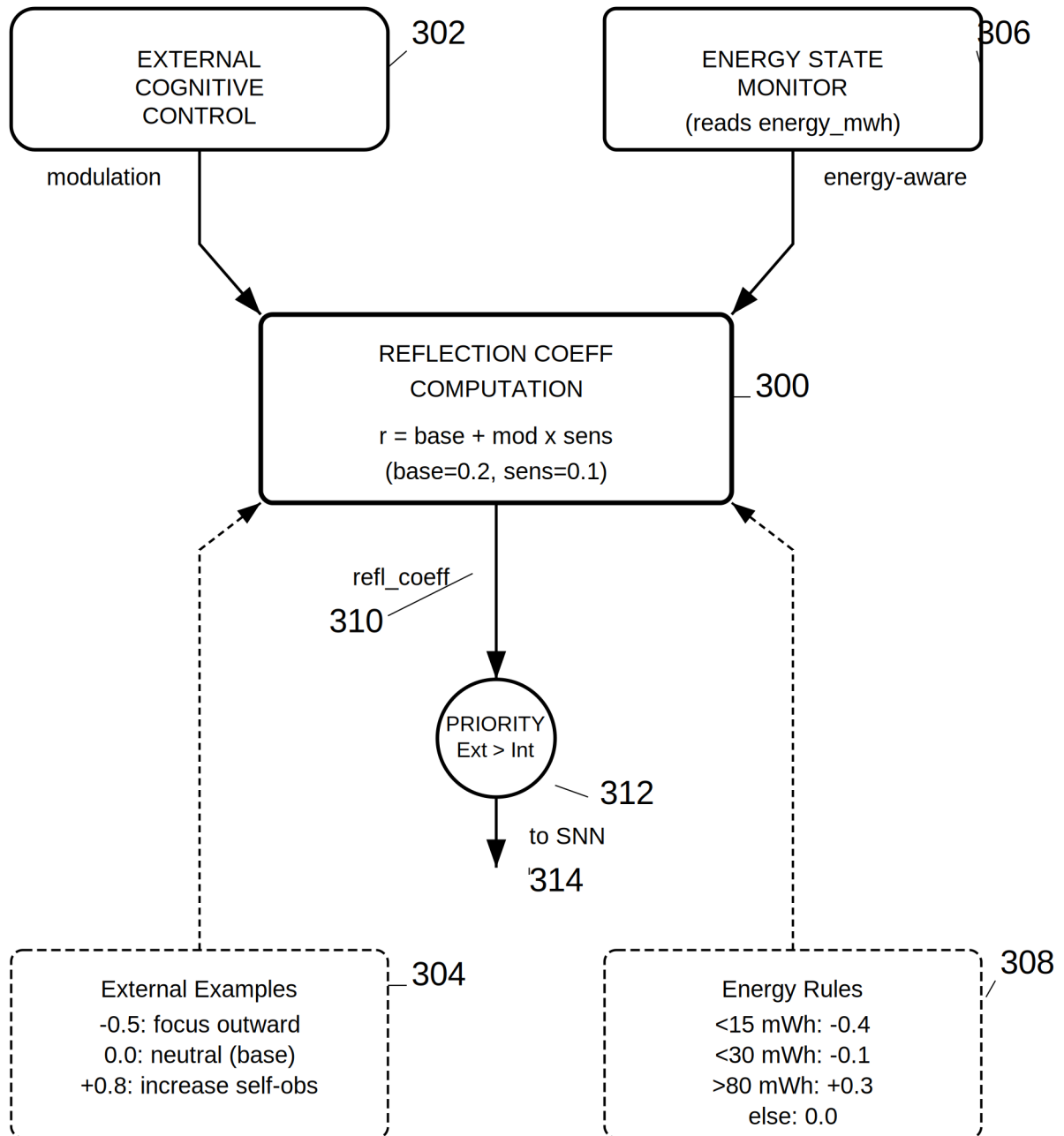


FIG. 3

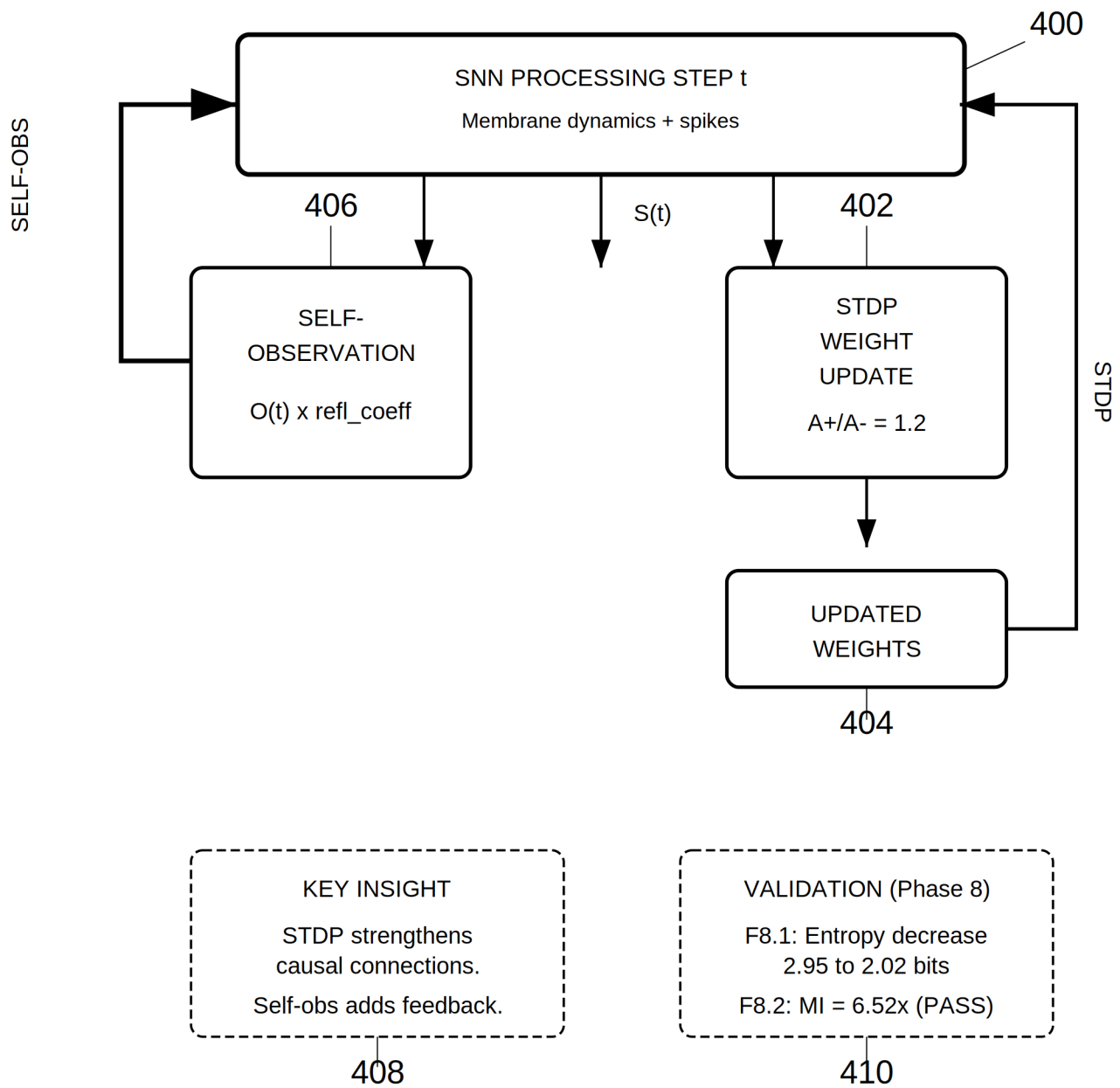


FIG. 4

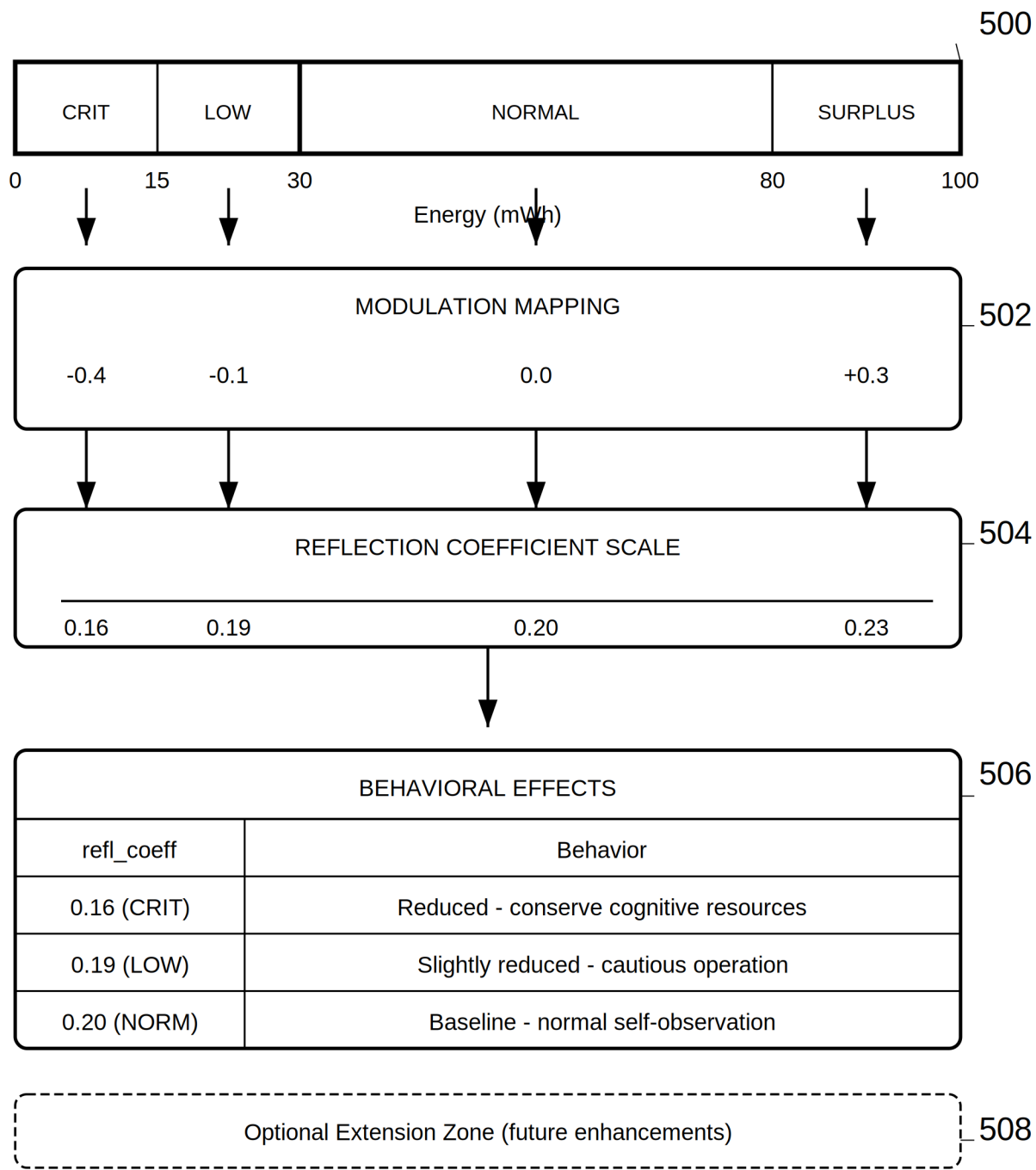
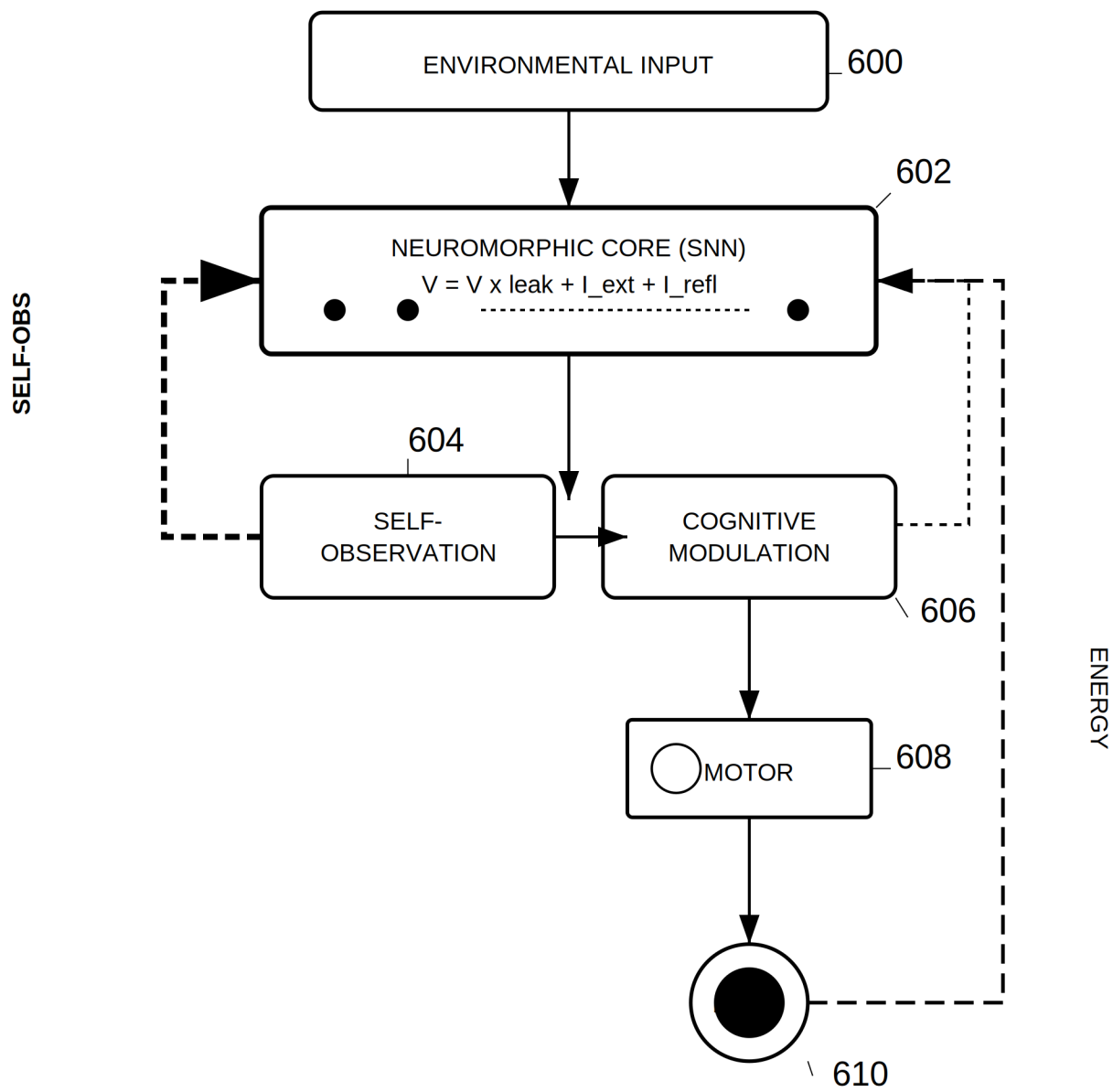


FIG. 5



Legend:

- Self-Observation Loop (Patent B - THIS PATENT)
- Energy Feedback Loop (Patent A)

FIG. 6

PATENT C

Autonomous Self-Regulation■and Cognitive Resynchronization in Neural Processing■Systems During Disconnection from

7 drawing sheets

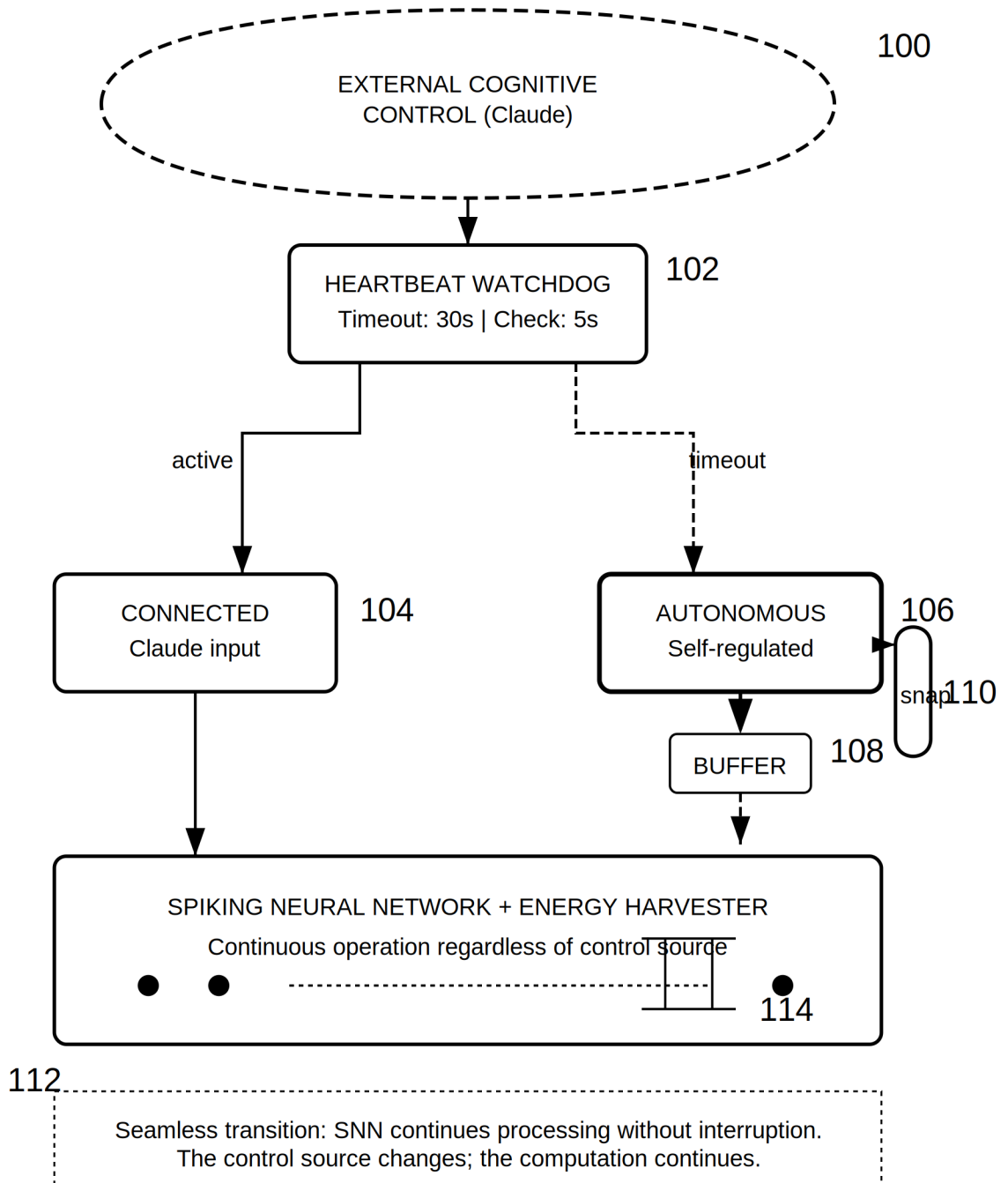


FIG. 1

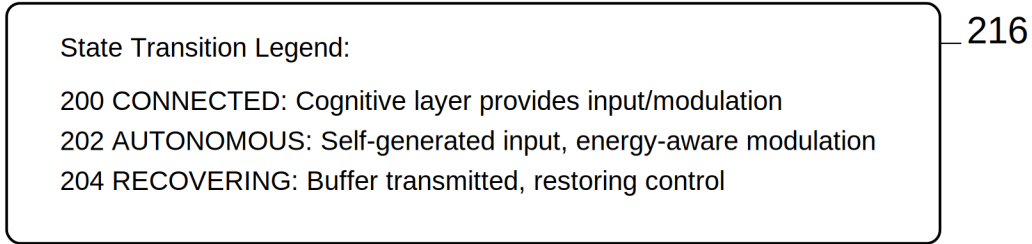
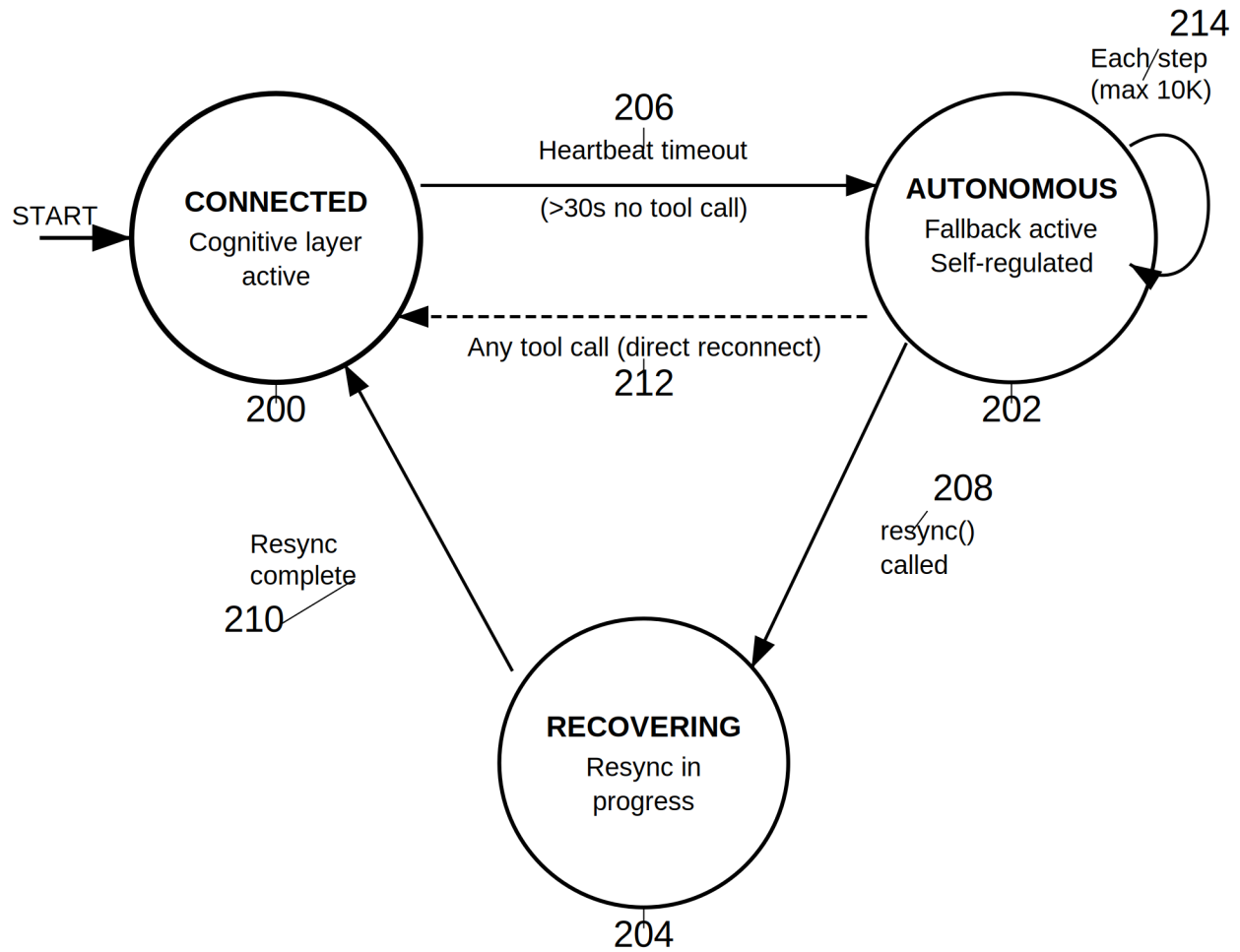
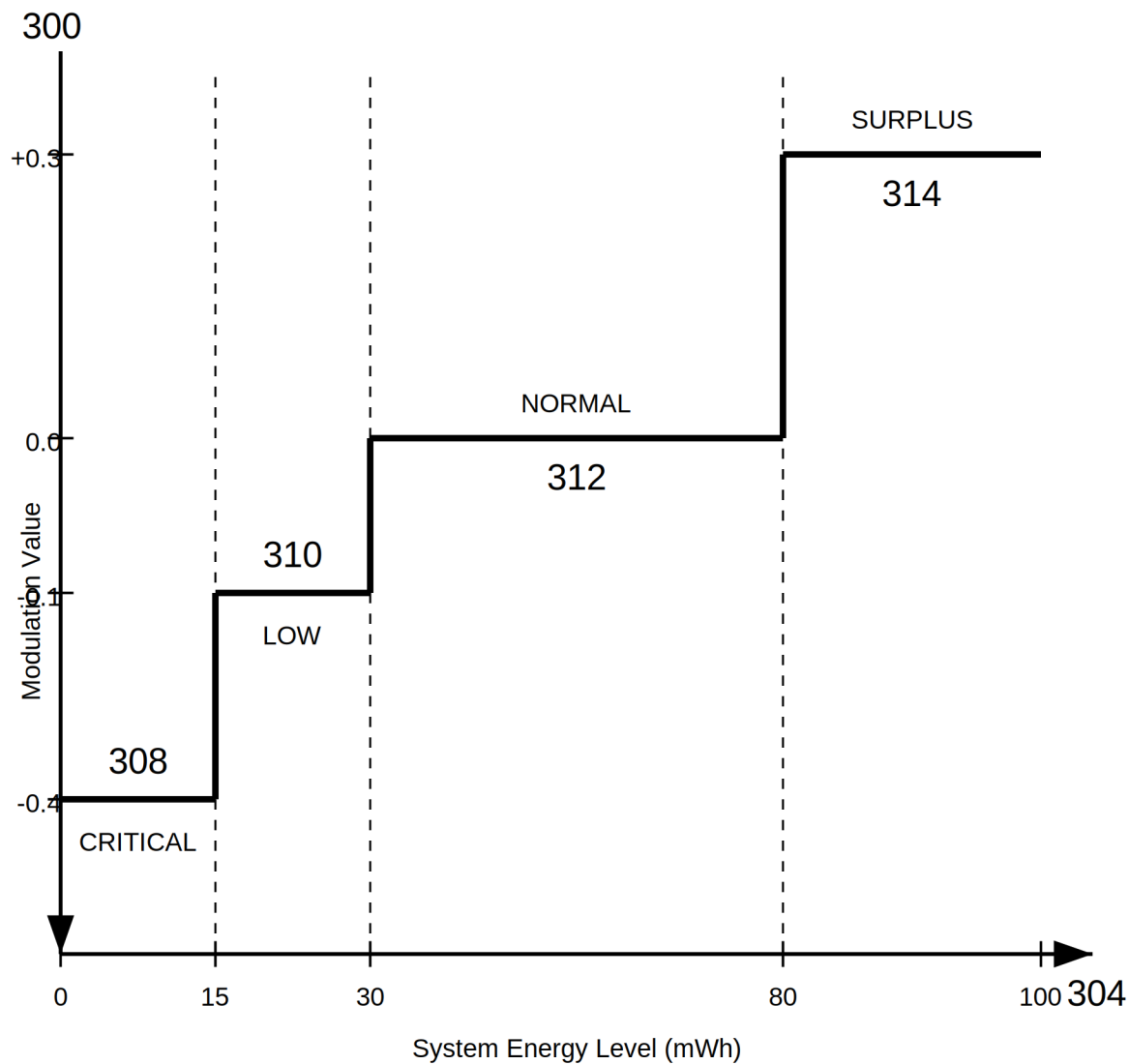


FIG. 2

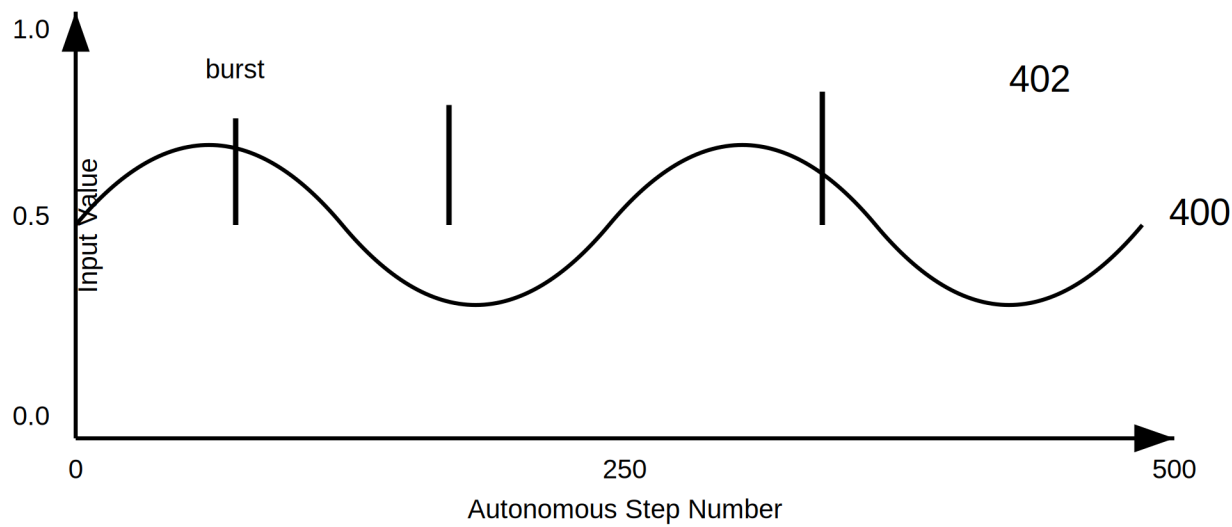


Behavioral Effects (302):

- CRITICAL (-0.4): Maximum conservation - inhibit neural activity
- LOW (-0.1): Cautious operation - slightly reduce activity
- NORMAL (0.0): Standard autonomous processing
- SURPLUS (+0.3): Opportunistic exploration - utilize excess

302

FIG. 3



$$\text{input} = 0.5 + 0.3 \times \sin(2 \times \pi \times \text{step} / \text{period}) + \text{burst}$$
$$\text{period} = 500 \text{ steps} \mid \text{burst probability} = 10\% \mid \text{burst amplitude} = +0.2$$

404

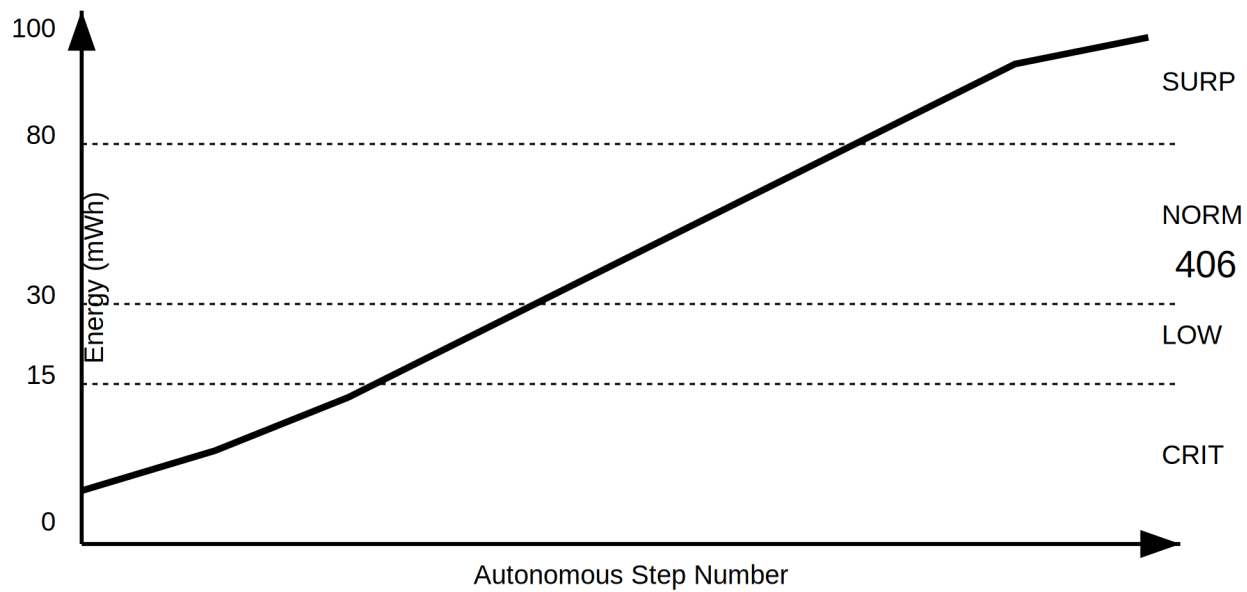


FIG. 4

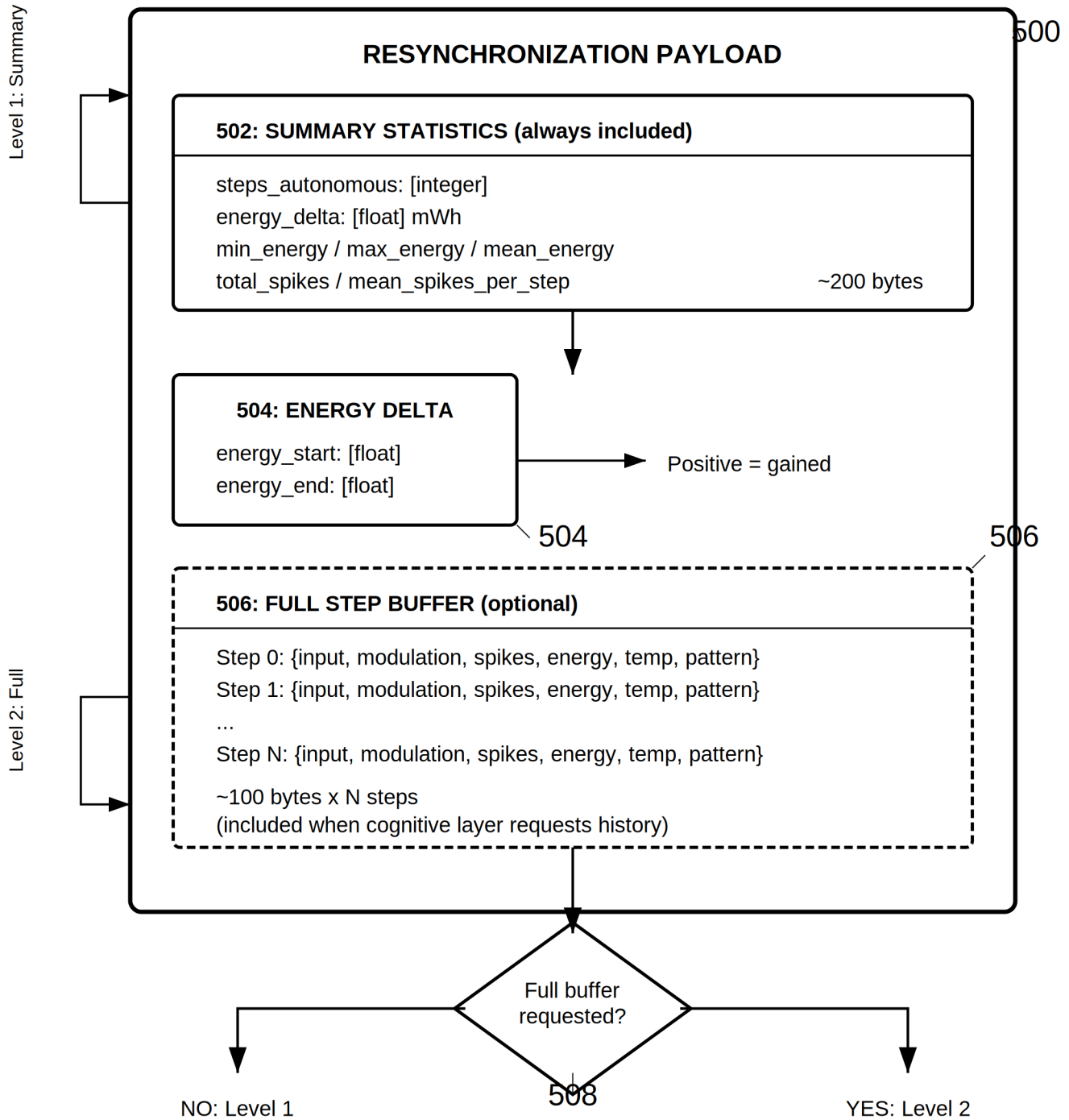
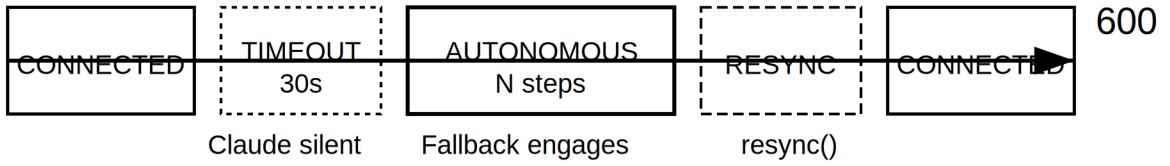
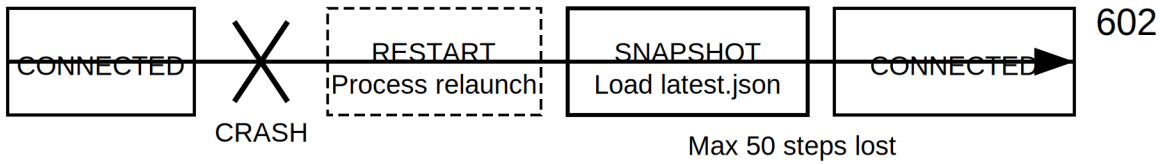


FIG. 5

600: NORMAL RECONNECTION



602: CRASH RECOVERY



604: CLEAN SHUTDOWN

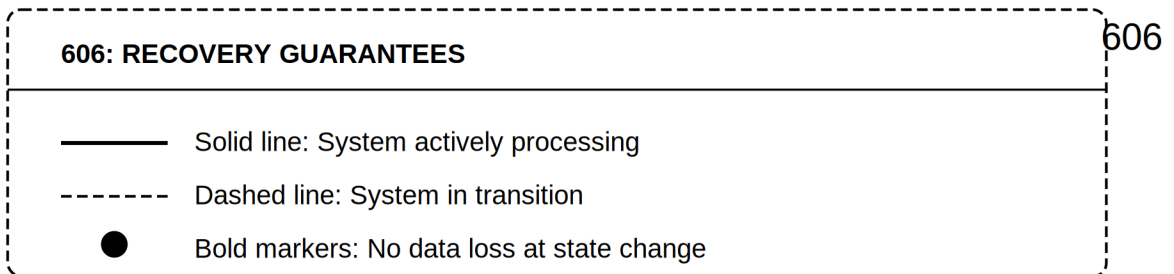
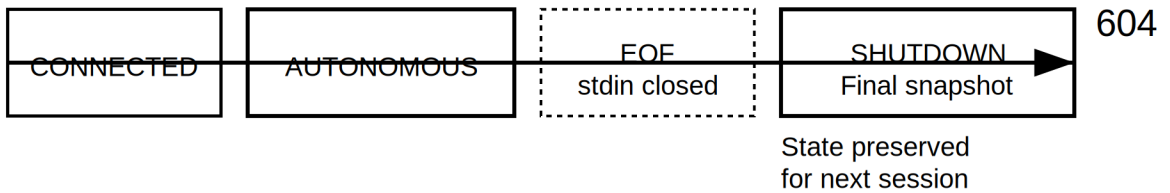
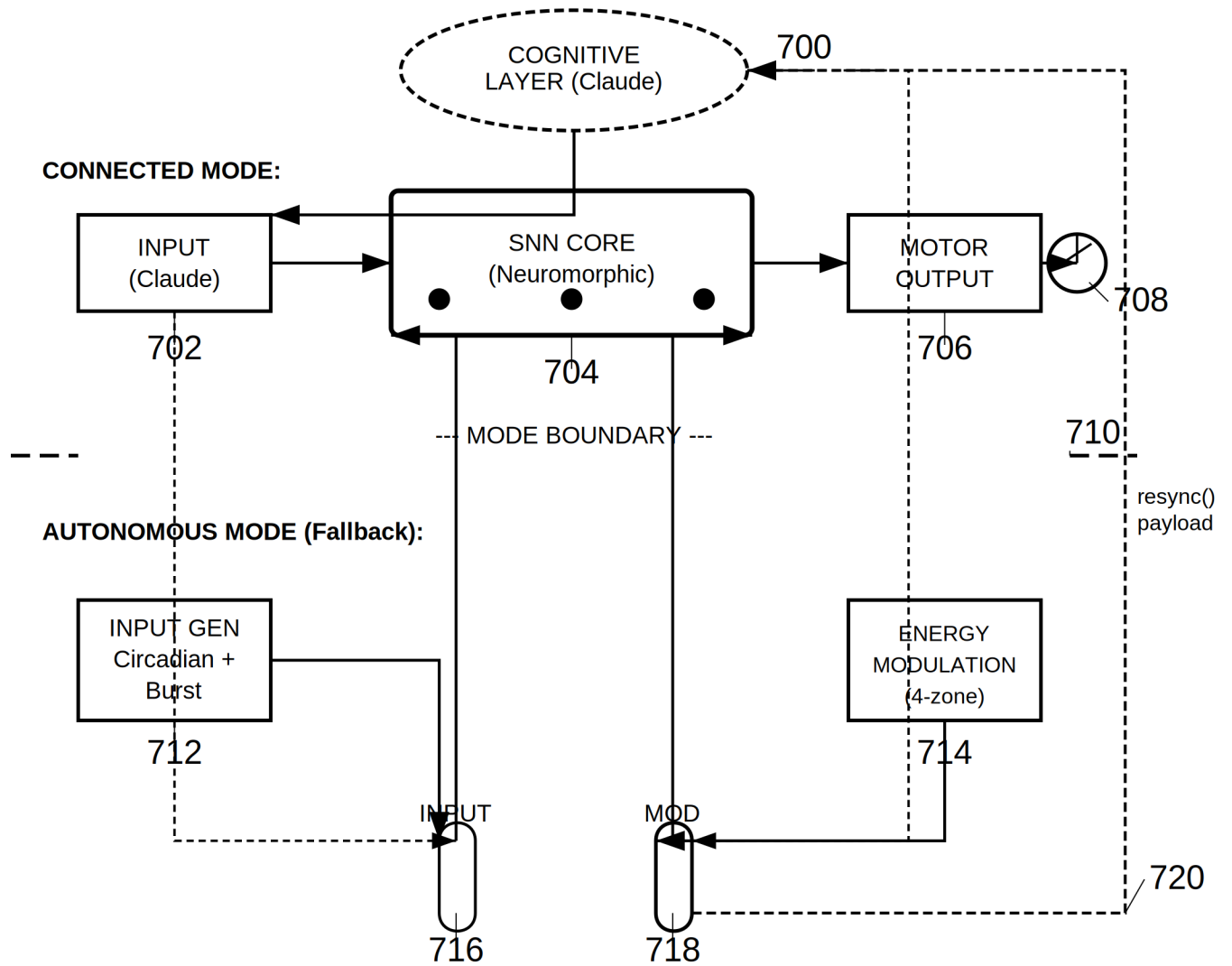


FIG. 6

END-TO-END SIGNAL FLOW: CONNECTED vs AUTONOMOUS



MODE SELECTION LOGIC:

CONNECTED: Input from Claude (702), modulation from Claude (700)
 AUTONOMOUS: Input from Generator (712), modulation from Energy (714)
 Switches (716, 718) select active source based on heartbeat status
 Feedback path (720) transmits resync payload on reconnection

FIG. 7